Assignment Plan - Blockchain-Based Medical Prescription Verification System

CCS6354 — Blockchain & Smart Contract Assignment

Group V | Topic Analysis Document

Blockchain-Based Medical Prescription Verification System

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1. Topic Overview

Field	Detail
Assignment Title	Blockchain-Based Medical Prescription Verification System
Group	Group V
Course	CCS6354 — Blockchain and Smart Contract
Submission Deadline	19 June 2026
Demo Weeks	Weeks 12–14
Blockchain Network	Ethereum — Sepolia Testnet
Dev Stack	Hardhat + Solidity ^0.8.x + ethers.js v5 + React.js
Money Involved?	No — pure verification system
Unique from other groups?	Yes — confirmed unique from all 24 registered groups

Core Value Proposition

> *"Current prescription systems — paper or electronic — are centralized, siloed, and easily forged. Our blockchain dApp replaces manual trust with on-chain verification: only admin-approved doctors can issue prescriptions, prescriptions are tamper-proof once issued, and pharmacies can verify and mark them as dispensed in real time — preventing forgery, double dispensing, and doctor shopping."*

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2. Team & Roles

Member	Role	Code Responsibility	Report Responsibility
Pharthiban	Group Leader, Report Lead, Presentation Lead	Core smart contracts, security patterns	Methodology, Literature Review
Derrel	Frontend — Logic & Integration	MetaMask connect, ethers.js integration, transaction logic, error handling	Discussion section
Dharven	Frontend — UI & Styling	UI pages, Tailwind/Bootstrap styling, `src/config/contract.js`	Abstract, Introduction, Conclusion, References (CITIC)
Ruben	Testing & Deployment	Automated test suite, gas reporter, Sepolia deployment, Etherscan verification	Results & Analysis section

> **Everyone** contributes to: system architecture diagram, demo video, and Q&A preparation.

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3. Problem Statement

What is Prescription Fraud?

Prescription fraud involves misusing medical prescriptions through:

- **Forgery** — creating fake prescriptions using stolen or fabricated prescription pads
- **Alteration** — changing dosage, quantity, or drug name on a legitimate prescription
- **Double dispensing** — presenting the same prescription at multiple pharmacies
- **Doctor shopping** — visiting multiple doctors to obtain multiple prescriptions for the same drug
- **Impersonation** — posing as a doctor or patient to obtain controlled substances
- **Expired prescription reuse** — presenting an outdated but previously valid prescription

Why It Matters

Prescription fraud is not a minor administrative problem — it is a life-and-death public health crisis:

- Fake medicines **kill patients** through toxic ingredients, zero active ingredients, or incorrect dosages
- Fraudulently obtained controlled substances (opioids, stimulants) fuel addiction epidemics
- Healthcare systems lose billions annually to fraudulent claims
- Developing countries are disproportionately impacted due to weaker verification infrastructure

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4. Global Problem Evidence

Scale of the Crisis

Statistic	Source
$432 billion — estimated annual value of counterfeit drug market, larger than arms, narcotics, and human trafficking combined	WHO, 2024
1 million deaths per year from fake/substandard medicines worldwide	WHO
267,000 deaths in sub-Saharan Africa from falsified antimalarial drugs annually	UNODC, 2023
169,000 deaths from fake antibiotics for childhood pneumonia (sub-Saharan Africa)	UNODC, 2023
10.5% of medicines in developing countries are substandard or counterfeit	WHO
1 in 2 medicines sold online globally is fake	WHO
19 million Americans likely obtained counterfeit medications through unverified online pharmacies	Kaiser Family Foundation
16 million Americans (6% aged 12+) abuse prescription drugs annually	NCDAS
Counterfeit incidents reported to WHO **doubled** in 2023 vs 2019	WHO
2,121 counterfeiting incidents in the US in 2022 alone, up 17% from prior year	Pharmaceutical Security Institute

Recent Real-World Cases (2024–2025)

- **January 2025**: WHO issued an active international medical alert for counterfeit cancer drug IMFINZI (durvalumab) found in Lebanon, Turkey, and Armenia — containing **zero active ingredients**, endangering cancer patients
- **2024**: Counterfeit HIV drugs containing antipsychotics instead of active ingredients distributed in the US supply chain
- **2023**: 15.5 million doses of illegally traded medicines seized globally in Operation Pangea (Interpol, 89 countries)
- **2023**: 72 arrests worldwide, 1,300 illegal pharmaceutical websites shut down (Operation Pangea XVI)

Prescription-Specific Fraud Statistics

- Approximately **85% of online pharmacies** offering controlled substances do not require a valid prescription
- **Doctor shopping**: Patients visit multiple doctors to accumulate prescriptions — primary driver of opioid crisis
- Drug overdoses from prescription opioids now **exceed those from cocaine and heroin combined**
- Paper prescriptions remain in use across most of Malaysia and developing Asia — easily forged

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5. Malaysian Context

Issue	Details
Legal framework	Under Malaysian law, selling counterfeit drugs carries up to **RM25,000 fine or 3 years imprisonment**, or both. Companies face up to RM50,000 fine.
Unregistered drugs	Malaysian Pharmacists Society (MPS) has flagged Malaysia's growing problem of **unregistered and adulterated** products — particularly prevalent in online channels
WHO risk category	Malaysia, as a middle-income developing country, falls in the WHO high-risk category where **10.5% of medicines** may be substandard or counterfeit
Digital health push	Ministry of Health Malaysia (KKM) has active digitization initiatives — blockchain prescription fits the national health IT direction
PDPA relevance	Malaysia's **Personal Data Protection Act (PDPA)** governs how medical data must be handled — relevant to our limitations discussion
University relevance	Widely used controlled substances like Ritalin/Adderall (study drugs) are a documented abuse concern in Malaysian university settings

6. Existing Solutions & Identified Gaps

Solution 1: Paper Prescriptions (current standard in Malaysia)

Problems:

- Easily forged using stolen prescription pads or computer-generated fakes
- Can be physically altered (dosage, quantity, drug name changed)
- No cross-pharmacy verification — same prescription can be filled at multiple pharmacies
- No real-time audit trail
- Human error prone — illegible handwriting, missing information

Solution 2: Existing Electronic Prescription Systems (EMR-based)

Problems:

- **Centralized architecture** — single point of failure, vulnerable to data breaches
- **Not interoperable** — different hospitals/clinics use incompatible systems
- Even Sweden's national e-prescribing system had significant issues: *23% of large-scale healthcare events were associated with e-prescribing system errors* (Rahman Jabin & Hammar, 2022)
- Does not prevent double dispensing across different pharmacy systems
- Not available in most of Malaysia — primary care clinics still use paper

Solution 3: Prescription Drug Monitoring Programs (PDMP)

Problems:

- State/country-level only — not interoperable globally or even nationally in many countries
- Not real-time — reporting delays mean fraud is detected after the fact
- Not implemented in Malaysia or most developing countries
- Requires law enforcement infrastructure to act on flagged data

The Gap Our System Fills

> Current solutions are either **too easy to forge** (paper), **too siloed** (EMR), or **too limited geographically** (PDMP). None provide a **decentralized, tamper-proof, real-time, interoperable** verification mechanism accessible to any pharmacy.

7. Does the Blockchain Solution Really Help?

Prescription Fraud Type	Current System	Our Blockchain Solution	Solved?
Forged paper prescription	Pharmacist manually checks handwriting/seal — easily fooled	Only on-chain prescriptions issued by admin-verified doctor wallets are valid	✅ Yes
Double dispensing (one Rx at 2 pharmacies)	No cross-pharmacy check	Smart contract marks prescription as DISPENSED — cannot be reused	✅ Yes
Doctor shopping (multiple doctors, same drug)	No shared database across clinics	All prescriptions on a shared immutable ledger — visible across pharmacies	✅ Yes
Altered prescriptions (dosage/quantity changed)	Paper can be physically modified after signing	On-chain hash of prescription is tamper-proof — any change invalidates it	✅ Yes

Prescription Fraud Type	Current System	Our Blockchain Solution	Solved?
Impersonation (fake doctor identity)	No real-time verification mechanism	Only admin-registered doctor wallets can issue prescriptions	✅ Yes
Expired prescription reuse	Manual date check — can be manipulated	Expiry timestamp encoded in smart contract — auto-rejects expired Rx	✅ Yes
Counterfeit online pharmacy drugs	Consumer has no way to verify drug origin	Prescription trail on-chain — drug supply chain is out of scope	⚠️ Partial
Rogue online pharmacies selling without Rx	Enforcement only — hard to track globally	Our system requires on-chain Rx to dispense — only for participating pharmacies	❌ Out of scope

Verdict: The core prescription fraud types — forgery, double dispensing, doctor shopping, and tampering — are directly and effectively solved. This is a genuine, meaningful solution to a life-critical problem.

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8. Proposed Solution Overview

How It Works (Step by Step)

```
Step 0: REGISTRATION PHASE (One-time setup)
        Doctor submits registration request on-chain
        (wallet address + MMC license number)
              ↓
        Pharmacy submits registration request on-chain
        (wallet address + Pharmacy Board registration number)
              ↓
        Admin (represents KKM/MMC) reviews pending requests
        Admin verifies license off-chain → approves or rejects on-chain
        DoctorApproved / PharmacyApproved events emitted
              ↓
Step 1: Doctor logs into the dApp with MetaMask (approved wallet only)
        Doctor issues a prescription (drug details, patient address, expiry)
        Prescription data is hashed (keccak256) — hash stored on-chain
        Actual data stored on IPFS (off-chain, for privacy)
        PrescriptionIssued event emitted
              ↓
Step 2: Patient receives their prescription ID
              ↓
Step 3: Pharmacist verifies prescription by ID (approved wallet only)
        Smart contract checks: valid? not expired? not yet dispensed?
        If all pass → pharmacist dispenses medication
        Contract marks prescription as DISPENSED
        PrescriptionDispensed event emitted
              ↓
Step 4: Prescription cannot be used again at any pharmacy
        Full audit trail permanently on-chain
```

Real-World Story for Demo

> *"A clinic submits a registration request on-chain with their Pharmacy Board number. The Admin (representing KKM) reviews and approves them. A doctor at the clinic similarly submits their MMC license number and gets approved. The doctor then issues an on-chain prescription for Ritalin to a patient. The patient goes to the approved pharmacy, shows their prescription ID. The pharmacist verifies on-chain — valid, not expired, not yet used — and dispenses the medication, marking it used. A second pharmacy attempt immediately fails — the contract rejects it. No forgery. No double dispensing. No doctor shopping. Zero ETH payments — purely trustless verification."*

Privacy Approach (IPFS + Hashing)

To protect patient medical data:

- **On-chain**: Only the `keccak256` hash of the prescription data + IPFS CID is stored
- **Off-chain (IPFS)**: Actual prescription data (drug name, patient name, dosage) stored encrypted
- This approach complies with Malaysia's PDPA by ensuring PII is not publicly exposed on-chain
- Follows best-practice "on-chain/off-chain hybrid" recommended by academic literature

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9. Registration Authority Design (Option B)

Who Is the Admin?

In the real world, two bodies govern medical practice in Malaysia:

Authority	Role in Our System
Malaysian Medical Council (MMC)	Issues and maintains doctor licenses — our Admin represents this body for doctor approvals
Pharmacy Board of Malaysia	Registers pharmacists and pharmacies — our Admin represents this body for pharmacy approvals
Ministry of Health Malaysia (KKM)	Oversees both — the deployer wallet represents KKM as the super admin

> In the dApp, the **Admin wallet = KKM/MMC delegated authority**. This is the real-world justification for the centralized admin role in our smart contract.

Registration Flow (Option B — Request + Approval)

```
DOCTOR REGISTRATION
─────────────────────────────────────────────
Doctor fills registration form on dApp:
  - Wallet address (auto-filled from MetaMask)
  - Full name
  - MMC Registration Number
  - Specialization
        ↓
Calls requestDoctorRegistration() on-chain
RegistrationRequested event emitted
        ↓
Admin sees pending request in Admin Panel
Admin verifies MMC number off-chain (via mmcmalaysia.com.my)
        ↓
If valid → Admin calls approveDoctor(requestId)
           DoctorApproved event emitted
           Doctor wallet is now DOCTOR_ROLE
If invalid → Admin calls rejectDoctor(requestId, reason)
             DoctorRejected event emitted

PHARMACY REGISTRATION
─────────────────────────────────────────────
Same flow but with Pharmacy Board registration number
requestPharmacyRegistration() → Admin reviews → approvePharmacy() / rejectPharmacy()
```

Why Option B Is Better for the Assignment

Aspect	Option A (Admin Direct)	Option B (Request + Approval)
Contract functions	2 fewer functions	More functions → better rubric marks
Events	Fewer events	More events → better rubric marks
Real-world accuracy	Admin has to know wallets upfront	Realistic — entities apply, authority approves
Demo flow	Less interesting	Shows full governance lifecycle
Access control complexity	Basic	Full RBAC with pending/approved/rejected states
Rubric B1 band	Meets	**Exceeds**

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10. Smart Contract Design

Contracts

Contract	Purpose
`PrescriptionRegistry.sol`	Core contract — manages registration requests, approvals, prescription lifecycle

Registration Structs & State

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.19;

enum RequestStatus { Pending, Approved, Rejected }

struct RegistrationRequest {
    address requester;
    string  name;
    string  licenseNumber;   // MMC no. (doctor) or Pharmacy Board no.
    string  ipfsCID;         // IPFS CID of supporting documents (optional)
    RequestStatus status;
    string  rejectionReason;
    uint256 submittedAt;
}

struct Prescription {
    address doctor;
    address patient;
    bytes32 dataHash;         // keccak256 of prescription data
    string  ipfsCID;          // IPFS CID of actual prescription data
    uint256 expiryTimestamp;
    bool    dispensed;
    bool    revoked;
    uint256 issuedAt;
    address dispensedBy;
}
```

Mappings

```
// Registration
mapping(uint256 => RegistrationRequest) public doctorRequests;
mapping(uint256 => RegistrationRequest) public pharmacyRequests;
uint256 public doctorRequestCount;
uint256 public pharmacyRequestCount;
```

```
// Approved roles (also managed via OpenZeppelin AccessControl)
mapping(address => bool) public verifiedDoctors;
mapping(address => bool) public verifiedPharmacies;

// Prescriptions
mapping(uint256 => Prescription) public prescriptions;
uint256 public prescriptionCount;

// Constants
uint256 public constant MAX_PRESCRIPTION_VALIDITY = 30 days;
```

Full Function List

Function	Access	Description
`requestDoctorRegistration(name, licenseNo, ipfsCID)`	Public	Doctor submits registration request on-chain
`requestPharmacyRegistration(name, licenseNo, ipfsCID)`	Public	Pharmacy submits registration request on-chain
`approveDoctor(requestId)`	Admin only	Approves doctor request, grants DOCTOR_ROLE
`rejectDoctor(requestId, reason)`	Admin only	Rejects doctor request with reason
`approvePharmacy(requestId)`	Admin only	Approves pharmacy request, grants PHARMACY_ROLE
`rejectPharmacy(requestId, reason)`	Admin only	Rejects pharmacy request with reason
`revokeDoctor(address)`	Admin only	Revokes a previously approved doctor
`revokePharmacy(address)`	Admin only	Revokes a previously approved pharmacy
`issuePrescription(patientAddr, dataHash, ipfsCID, expiry)`	Doctor only	Issues prescription, stores hash + IPFS CID
`getPrescription(prescriptionId)`	Public (READ)	Returns prescription details by ID
`verifyPrescription(prescriptionId)`	Pharmacist only	Checks: valid, not expired, not dispensed
`dispensePrescription(prescriptionId)`	Pharmacist only	Marks as dispensed — irreversible
`revokePrescription(prescriptionId)`	Doctor / Admin	Cancels prescription (wrong dosage etc.)
`getPendingDoctorRequests()`	Admin only (READ)	Returns all pending doctor requests
`getPendingPharmacyRequests()`	Admin only (READ)	Returns all pending pharmacy requests

Events (Well Exceeds the 2-Event Minimum)

```
// Registration events
event DoctorRegistrationRequested(uint256 indexed requestId, address indexed requester, string licenseNumber);
event DoctorApproved(uint256 indexed requestId, address indexed doctor);
event DoctorRejected(uint256 indexed requestId, address indexed requester, string reason);
event PharmacyRegistrationRequested(uint256 indexed requestId, address indexed requester, string licenseNumber);
event PharmacyApproved(uint256 indexed requestId, address indexed pharmacy);
event PharmacyRejected(uint256 indexed requestId, address indexed requester, string reason);
event DoctorRevoked(address indexed doctor);
event PharmacyRevoked(address indexed pharmacy);
```

```
// Prescription events
event PrescriptionIssued(uint256 indexed prescriptionId, address indexed doctor, address indexed patient,
uint256 expiry);
event PrescriptionDispensed(uint256 indexed prescriptionId, address indexed pharmacy, uint256 timestamp);
event PrescriptionRevoked(uint256 indexed prescriptionId, address indexed revokedBy);
```

Security Patterns (Rubric B1 — Exceeds Band)

```
// SECURITY: Checks-Effects-Interactions on dispensePrescription
function dispensePrescription(uint256 _id) external onlyRole(PHARMACY_ROLE) {
    // CHECKS
    require(prescriptions[_id].issuedAt != 0, "Prescription does not exist");
    require(!prescriptions[_id].dispensed, "Already dispensed");
    require(!prescriptions[_id].revoked, "Prescription has been revoked");
    require(block.timestamp <= prescriptions[_id].expiryTimestamp, "Prescription has expired");
    // EFFECTS (state change BEFORE any external interaction)
    prescriptions[_id].dispensed = true;
    prescriptions[_id].dispensedBy = msg.sender;
    // INTERACTIONS
    emit PrescriptionDispensed(_id, msg.sender, block.timestamp);
}

// SECURITY: Role-Based Access Control (OpenZeppelin AccessControl)
bytes32 public constant DOCTOR_ROLE   = keccak256("DOCTOR_ROLE");
bytes32 public constant PHARMACY_ROLE = keccak256("PHARMACY_ROLE");
// DEFAULT_ADMIN_ROLE used for KKM/MMC admin

// SECURITY: Input validation on all write functions
function requestDoctorRegistration(
    string calldata _name,           // calldata for gas optimisation
    string calldata _licenseNumber,
    string calldata _ipfsCID
) external {
    require(bytes(_name).length > 0, "Name cannot be empty");
    require(bytes(_licenseNumber).length > 0, "License number required");
    require(!verifiedDoctors[msg.sender], "Already a verified doctor");
    // ... registration logic
}
```

Gas Optimisation Notes

- Use `calldata` instead of `memory` for all read-only string parameters
- Use `events` for audit logs instead of storing history arrays on-chain
- Mark `getPrescription()` and `getPendingRequests()` as `view`
- Named constants for all magic values (`MAX_PRESCRIPTION_VALIDITY` , role hashes)
- Struct packing: `bool dispensed` and `bool revoked` packed together in storage slot

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11. System Architecture

```
┌──────────────────────────────────────────────────────────┐
│                        USER LAYER                        │
│   Doctor (MetaMask)  │  Patient  │  Pharmacist (MM)      │
└──────────────────────────────────────────────────────────┘
               │                            │
               ▼                            ▼
┌──────────────────────────────────────────────────────────┐
```

```
┌────────────────────────────────────────────────────┐
│                FRONTEND LAYER                      │
│          React.js App (Derrel + Dharven)           │
└────────────────────────────────────────────────────┘
                          │
                          ▼
┌────────────────────────────────────────────────────┐
│              INTEGRATION LAYER                     │
│                ethers.js v5                        │
└────────────────────────────────────────────────────┘
                          │
                          ▼
┌────────────────────────────────────────────────────┐
│          SMART CONTRACT LAYER (Pharthiban)         │
│     PrescriptionRegistry.sol  (Solidity ^0.8)      │
│     OpenZeppelin: AccessControl + Ownable          │
└────────────────────────────────────────────────────┘
              │                            │
              ▼                            ▼
┌────────────────────┐      ┌───────────────────────────┐
│  BLOCKCHAIN LAYER  │      │       STORAGE LAYER       │
│  Sepolia Testnet   │      │      IPFS via Pinata      │
│  (Ruben deploys)   │      │  (Prescription data CIDs) │
└────────────────────┘      └───────────────────────────┘
```

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12. UI Plan (6 Pages)

Page 1 — Home / Landing

- Hero section: what the dApp solves (prescription fraud problem + WHO statistics)
- "Connect Wallet" button (MetaMask)
- Network warning banner if not on Sepolia (chain ID: 11155111)
- Role detection after connect: redirects to correct dashboard based on wallet role
- "Register as Doctor" and "Register as Pharmacy" buttons for unregistered wallets

Page 2 — Admin Panel (Admin wallet only)

> **Real-world authority**: This page is accessed only by the KKM/MMC admin wallet (contract deployer).

Doctor Requests Tab:

- Table of pending doctor registration requests: name, MMC license no., wallet address, submitted date
- Approve button → calls `approveDoctor(requestId)` → MetaMask confirmation
- Reject button → modal to enter rejection reason → calls `rejectDoctor(requestId, reason)`
- Tabs: Pending / Approved / Rejected — with status badges

Pharmacy Requests Tab:

- Same layout for pharmacy requests
- Approve / Reject with Pharmacy Board registration number shown

Revocation Tab:

- List of currently verified doctors and pharmacies
- Revoke button per entry (for misconduct or license expiry)

Page 3 — Registration Request Page (Public — Doctors & Pharmacies)

- Two tabs: "Register as Doctor" / "Register as Pharmacy"
- **Doctor form**: wallet address (auto-filled), full name, MMC registration number, specialization, upload supporting document (to IPFS)
- **Pharmacy form**: wallet address (auto-filled), pharmacy name, Pharmacy Board registration number, address, upload license document (to IPFS)
- Submit → `requestDoctorRegistration()` / `requestPharmacyRegistration()` → MetaMask → pending status shown
- Status tracker: Pending → Approved / Rejected (with reason if rejected)

Page 4 — Doctor Dashboard (Approved Doctor wallet only)

- Summary cards: total prescriptions issued, active, dispensed, revoked
- "Issue Prescription" form:
  - Patient wallet address input
  - Drug details input (sent to IPFS — not stored raw on-chain)
  - Dosage, frequency, duration
  - Expiry date picker (max 30 days)
  - Submit → MetaMask → transaction status → prescription ID shown on success
- My Prescriptions table: ID, patient, issued date, expiry, status badge

Page 5 — Pharmacist Dashboard (Approved Pharmacy wallet only)

- Search by Prescription ID input
- Verify button (READ — `verifyPrescription()`) → shows:
  - Valid / Expired / Already Dispensed / Revoked — with colour-coded badges
  - Prescription details: doctor name, drug hash, expiry
- If valid: "Dispense" button → calls `dispensePrescription(prescriptionId)` → MetaMask → confirmed
- Cannot dispense twice — contract rejects and UI shows user-readable message: *"This prescription has already been dispensed."*
- Dispensing history table

Page 6 — Patient View (Any connected wallet)

- Lists all prescriptions issued to the connected wallet address
- Shows: prescription ID, issuing doctor wallet, issued date, expiry date, status (Active / Dispensed / Revoked / Expired)
- QR code generated for each active prescription (encoding the prescription ID for pharmacist to scan)
- Copyable prescription ID for easy sharing

Shared Requirements (All Pages)

- `src/config/contract.js` — ABI + contract address stored here, not hardcoded in components
- Transaction status visible on all write actions: Pending → Confirmed → Failed
- All Solidity revert messages translated to user-readable text
- Responsive layout with Tailwind CSS
- README.md with full setup and run instructions

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13. Limitations & How to Frame Them

Limitation	Why it exists	How to handle
Medical data privacy on public chain	Blockchain is public — prescription details visible to all	**Fixed in design**: Only `keccak256` hash + IPFS CID on-chain. Actual data stored on

Limitation	Why it exists	How to handle
		IPFS off-chain
GDPR/HIPAA/PDPA — right to be forgotten	Blockchain immutability conflicts with data deletion laws	Acknowledge as limitation. Future work: permissioned blockchain (Hyperledger). Our scope: no PII on-chain
Admin is still a centralized trust point	A single admin wallet approves all doctors/pharmacies — if compromised, rogue approvals possible	Acknowledge as limitation. Future work: multi-sig admin (2-of-3 KKM signers required to approve)
Doctor's wallet key compromised	Stolen private key = ability to issue fake prescriptions	Limitation. Future work: multi-sig wallets + admin revocation mechanism (already included in contract)
Network adoption dependency	All pharmacies need blockchain access to verify	Acknowledge. Future work: QR code gateway for non-blockchain pharmacies
Gas fees per prescription	Each on-chain transaction costs ETH	Limitation for production. Mitigated by Sepolia (free) for demo. Future work: Layer 2 (Polygon)
Offline/rural pharmacies cannot verify	No internet or blockchain access in rural areas	Acknowledge. Our scope: urban digital pharmacies
Does not fix drug supply chain counterfeiting	Counterfeit drugs enter supply chain before prescription	Clearly scoped out. "We solve the PRESCRIPTION FRAUD problem, not the upstream supply chain problem"

Scope Statement for Report

> *"This system addresses the last-mile prescription verification problem: preventing fraudulent issuance, alteration, and reuse of prescriptions at the doctor-to-pharmacy level. It does not address upstream pharmaceutical supply chain counterfeiting, which requires separate infrastructure beyond the scope of this work."*

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14. Academic Papers (12+ Confirmed)

All papers are post-2020 and from peer-reviewed journals or conferences.

#	Paper	Authors	Year	Journal/Source	Relevance
1	Use of Blockchain Technology for Electronic Prescriptions	Seaberg, Seaberg, Seaberg	2021	PMC / BHTY Journal	Direct topic match — blockchain e-prescribing
2	SVBE: Searchable and Verifiable Blockchain-based EMR System	Alrebdi et al.	2022	Nature Scientific Reports	High-impact — blockchain EMR with IPFS
3	Harmonizing sensitive data exchange and double-spending prevention	Schlatt, Sedlmeir et al.	2022	arXiv / ACM	E-prescription + double-spending prevention
4	Blockchain and Raft consensus for secure physician prescriptions	Abdorrahimi et al.	2024	Nature Scientific Reports	Prescriptions + consensus mechanisms
5	Towards an Optimized Blockchain-Based Secure Medical Prescription-Management System	—	2024	MDPI Future Internet	Solidity + Truffle implementation — directly applicable
6	A Blockchain Based Approach for E-Prescription System Security and Traceability	—	2023	CEUR-WS	IPFS + smart contracts for prescriptions

#	Paper	Authors	Year	Journal/Source	Relevance
7	Electronic prescription system requirements: a scoping review	—	2022	PMC	Gaps in current e-prescription systems
8	Issues with the Swedish e-prescribing system	Rahman Jabin, Hammar	2022	PMC / SAGE	Proves existing EMR e-prescribing systems fail
9	Challenges and advantages of electronic prescribing system	—	2023	PMC	Survey study on e-prescribing limitations
10	Blockchain-Based Healthcare Records Management Framework	—	2024	MDPI Technologies	Security, privacy, interoperability in blockchain healthcare
11	Advancing Compliance with HIPAA and GDPR in Healthcare via Blockchain	—	2025	PMC	PDPA/GDPR framing for limitations section
12	Privacy preservation in blockchain-based healthcare data sharing	—	2025	Springer	Systematic review — on-chain/off-chain mechanisms

Supporting Primary Sources (for problem statement)

- WHO Substandard and Falsified Medical Products Reports (2022–2025)
- UNODC Report on counterfeit medicines in sub-Saharan Africa (2023)
- Pharmaceutical Security Institute Incident Trends (2023)
- Operation Pangea XVI Interpol Report (2023)

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15. Rubric Fit Analysis

Section A — Research Report (50 marks)

Criterion	Why This Topic Scores High	Expected Band
A1 Abstract & Introduction	Life-or-death problem with $432B global market, 1M deaths/year — compelling urgency. Clear objectives.	**Exceeds**
A2 Literature Review	12+ post-2020 peer-reviewed papers confirmed. Clear gap: existing e-prescribing is centralized and siloed.	**Exceeds**
A3 Methodology	Multi-role RBAC, IPFS hybrid storage, Request+Approval governance, full system architecture with trade-off analysis	**Exceeds**
A4 Results & Analysis	Clear test scenarios: register → approve → issue → verify → dispense → reject reuse. Gas table from hardhat-gas-reporter.	**Meets/Exceeds**
A5 Conclusion & Future Work	Realistic future work: multi-sig admin, PDPA compliance, Layer 2, QR gateway for offline pharmacies	**Exceeds**

Section B — Development Artefact (40 marks)

Criterion	Why This Topic Scores High	Expected Band
B1 Smart Contract	RBAC (3 roles), 2 registration structs, 11+ mappings, 11 events, 15 functions, CEI pattern, AccessControl, enum for request status	**Exceeds**
B2 Deployment	Hardhat + Sepolia testnet + Etherscan verification + ABI + deployment script + README	**Exceeds**

Criterion	Why This Topic Scores High	Expected Band
B3 Testing	3 roles × request/approve/reject/issue/dispense = very rich test matrix. Multi-account simulation natural.	**Exceeds**
B4 Frontend	6 pages, 4 distinct user flows (Admin/Doctor/Pharmacist/Patient), MetaMask, tx status, error handling, config.js	**Exceeds**

Section C — Presentation (10 marks)

Criterion	Why This Topic Scores High	Expected Band
Slide Quality	Compelling narrative: WHO data → problem → solution → governance model → demo.	**Exceeds**
Live Demo	Register doctor → Admin approves → issue Rx → verify → dispense → attempt reuse (rejected). Full 6-step flow on Sepolia.	**Exceeds**
Q&A	Every design decision (IPFS, RBAC, Request+Approval, CEI) has clear rationale tied to real-world authority (MMC/KKM).	**Exceeds**

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16. Development Stack

Component	Technology	Notes
Smart Contract Language	Solidity ^0.8.x	Required — no older versions
Blockchain Network	Ethereum Sepolia Testnet	Preferred — highest demo marks
IDE	Remix IDE / VS Code + Solidity extension	—
Deployment Framework	**Hardhat**	Chosen for Hardhat gas-reporter + TypeScript support
Local Blockchain	Ganache GUI v2.x (port 7545)	For early testing before Sepolia
Wallet	MetaMask browser extension	Connected to Sepolia
Frontend	React.js	Vite or Create React App
Blockchain Library	ethers.js v5	Recommended over web3.js
Styling	Tailwind CSS	Clean, responsive
Decentralized Storage	IPFS via Pinata	For prescription data (off-chain)
Testing Framework	Hardhat test (Chai/Waffle)	With hardhat-gas-reporter
Libraries	OpenZeppelin (AccessControl, Ownable)	For RBAC and security patterns

Environment Configuration

```
.env file (gitignored):
SEPOLIA_RPC_URL=https://sepolia.infura.io/v3/YOUR_KEY
DEPLOYER_PRIVATE_KEY=0x...
ETHERSCAN_API_KEY=...
```

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17. Topic Comparison (Why This Over Others)

Topics Considered & Rejected

Topic	Key Reason for Rejection
Decentralized Royalty Distribution (original)	ETH payments core to use case — ETH volatility, complex security patterns, many limitations
NFT Event Ticketing	Money unavoidably core; scalping problem not truly solved by NFTs; fewer academic papers
Student Attendance System	Problem is mild — biometrics already solve it better and cheaper; gas fees per attendance impractical

Why Medical Prescription Wins

Factor	Medical Prescription
Problem severity	Life-or-death — WHO, 1M deaths/year
Money involved	None — pure verification
Does solution work	Yes — directly prevents all major fraud types
Academic papers	12+ confirmed post-2020
Demo clarity	4-step crystal clear flow
Malaysian relevance	KKM digitization, PDPA, MPS warnings
Unique from other groups	Yes — confirmed unique
Smart contract complexity	Very rich — 3 roles, 15 functions, 11 events, IPFS, RBAC, Request+Approval governance
Rubric alignment	Maps to Exceeds band across all criteria

Document prepared by Pharthiban (Group Leader) — CCS6354 Group V Last updated: May 2026 — v2.0 (Registration Authority: Option B — Request + Approval)